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Top 100 Orthopedic AI • Rank #63

Personalized Medical Recommendation System Using Collaborative Filtering

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CLINICAL & AGENTIC AI TAKEAWAY

This personalized medical recommendation system using collaborative filtering and supervised machine learning algorithms helps clinicians select effective treatment plans by suggesting treatment options based on similar past cases.

ABSTRACT

The complexity of patient information and inconsistencies in treatment results are becoming a challenge to clinical decision-making in healthcare. This paper introduces an individualized medical recommendation system based on collaborative filtering and supervised machine learning algorithms to help clinicians select effective treatment plans. Anonymized clinical data, including demographic features, diagnostic codes, lab results, and treatment outcomes, is analyzed to compute patient similarity based on clinically weighted similarity measures. A combined collaborative filtering system is used to create treatment suggestions using effective results that have been seen in related patient profiles. Parallel to this, the prediction of the disease through the use of supervised classification models is used to enhance the accuracy of the diagnosis. The suggested system makes recommendations explainable as the suggestions are associated with evidence of similar cases in history. It is experimentally demonstrated that the disease prediction model has an accuracy of 96% with the help of a Support Vector Classifier and the recommendation module has F1-scores of up to 0.50. The suggested system is used to support clinical decisionmaking as it suggests treatment options based on the results of clinically comparable past occurrences.

Study Overview & Background

The complexity of patient information and inconsistencies in treatment results are becoming a challenge to clinical decision-making in healthcare. This paper introduces an individualized medical recommendation system based on collaborative filtering and supervised machine learning algorithms to help clinicians select effective treatment plans. Anonymized clinical data, including demographic features, diagnostic codes, lab results, and treatment outcomes, is analyzed to compute patient similarity based on clinically weighted similarity measures. A combined collaborative filtering system is used to create treatment suggestions using effective results that have been seen in related patient profiles. Parallel to this, the prediction of the disease through the use of supervised classification models is used to enhance the accuracy of the diagnosis. The suggested system makes recommendations explainable as the suggestions are associated with evidence of similar cases in history. It is experimentally demonstrated that the disease prediction model has an accuracy of 96% with

the help of a Support Vector Classifier and the recommendation module has F1-scores of up to 0.50. The suggested system is used to support clinical decisionmaking as it suggests treatment options based on the results of clinically comparable past occurrences.

Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

This personalized medical recommendation system using collaborative filtering and supervised machine learning algorithms helps clinicians select effective treatment plans by suggesting treatment options based on similar past cases.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** P. S, Maturu Thrissaptha Reddy, Ethakota Sruthi, Bhagath Kalyani, Mohamed Sithik M, R. S
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